STOCHASTIC METHODS IN WATER RESOURCES

Unit 3: Random functions

Lecture 9: Definitions, random function types, stationary random functions

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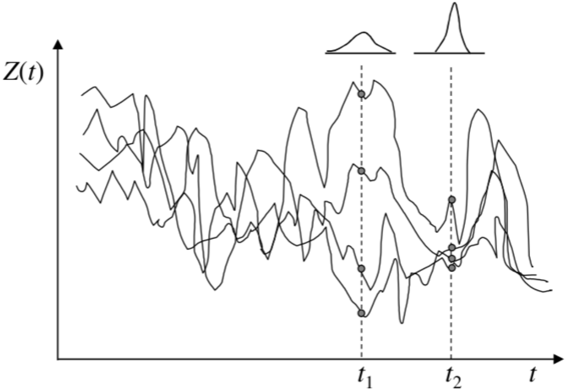
Definitions

- Consider a hydrological variable Z that can either varies in space (e.g. hydraulic conductivity) or in time (e.g. surface water level) or in space and time (e.g. growndwater depth).
- The value of any of these variables is known only in those places where it is observed, and it is unknown in other places.
- Accordingly, the value of Z is uncertain, to deal with this, instead of using one single functio either in time ($Z(t)$), or space ($Z(\mathbf{x})$) or space and time ($Z(\mathbf{x}, t)$), multiple functions are used (see figure for $Z(t)$).
- Each of these functions have the same probability of representing the true.
- The family of equally probable functions representing Z is known as an **emsemble** or a **random function**.



Definitions

- **Random field**: If Z is a function of space $(\mathbf{x})$, $Z(\mathbf{x})$, $\mathbf{x} = (x, y, z)$.
- **Random or stochastic process**: If Z is a function of time (t), $Z(t)$.
- **Space-time random field**: If Z is a function of space $(\mathbf{x})$ and time (t), $Z(\mathbf{x}, t)$, $\mathbf{x} = (x, y, z)$.

Here, we are going to focus on random process $(Z(t))$. Note that in the figure above, four functions $Z(t)$ (ensemble members) are shown and each of them are know as realizations and denoted as $z(t)$. The number of realizations, depending on the variable's nature, can be finite or infinite.
Another way of explaining this concept is to imagine a random function as a collection of random variables (one at every location in space or point in time). Observing at the figure above, for each time t, a $Z(t)$ is defined. This is why for each t, $Z(t)$ is described with a pdf $f_Z(z; t)$, which depends not only on the value of z but also on the value of t. These pdf is estimated based on sample of data taken at particular time (e.g. t_1). As shown in the figure, the pdfs of the data sample at t_1 and t_2 are different, and indicate that variance is larger in t_1 than in t_2 and thus more uncertain.
At each point in time (t), μ_Z is calculated as:

$$\mu_Z(t) = E[Z(t)] = \int_{-\infty}^{\infty} z f_Z(z; t) dz$$

and σ_Z^2 is:

$$\sigma_Z^2(t) = E[(Z(t) - \mu_Z(t))^2] = \int_{-\infty}^{\infty} (z - \mu_Z(t))^2 f_Z(z; t) dz$$

Definitions

The random variables are usually correlated in time (or in space for spatial random functions). This means that the covariance $COV[Z(t_1), Z(t_2)] \neq 0$. Note that the further apart the points (e.g. t_1 and t_2) the lower the covariance. This is estimated as

$$\begin{aligned} COV[Z(t_1), Z(t_2)] &= E[(Z(t_1) - \mu_Z(t_1))(Z(t_2) - \mu_Z(t_2))] \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (z_1 - \mu_Z(t_1))(z_2 - \mu_Z(t_2)) f_Z(z_1, z_2; t_1, t_2) dz_1 dz_2 \end{aligned}$$

if $t_1 = t_2$, $COV[Z(t_1), Z(t_2)] = \sigma_Z^2(t)$.
For n different locations t_i where $i = 1, 2, \cdots, n$, the probability measure that a realization of $Z(t)$ has the values between $z_1 < z(t_1) \leq z_1 + dz_1$, $z_2 < z(t_2) \leq z_2 + dz_2$, $\cdots$, $z_n < z(t_n) \leq z_n + dz_n$. The associated multivariate pdf $f_{Z(t_1), Z(t_2), \cdots, Z(t_n)}(z_1, z_2, \cdots, z_n; t_1, t_2, \cdots, t_n)$

$$f_{Z(t_1), Z(t_2), \cdots, Z(t_n)}(z_1, z_2, \cdots, z_n; t_1, t_2, \cdots, t_n) = \lim_{\partial z_1, \cdots, \partial z_n \to 0} \frac{Pr(z_1 < z(t_1) \leq z_1 + \partial z_1, z_2 < z(t_2) \leq z_2 + \partial z_2, \cdots, z_n < z(t_n) \leq z_n + \partial z_n)}{\partial z_1 \partial z_2 \cdots \partial z_n}$$

The multivariate pdf of a random function is sometime referred to as **multipoint pdf**, because it relates to random variables at different points or locations. Note that random function is fully characterized if the multivariate pdf for any set of points is known.

Types of random functions